Supplementary Material

Algorithms and Additional Results for the Paper "*How Low Can You Go? The Data-Light SE Challenge*"

Overview

This supplementary document contains full algorithm listings and large-format tables referenced in the main paper. These materials are hosted online for readability (e.g., GitHub) but are provided here in a unified PDF.

Algorithm 1: LINE — Distance^2 Seeding with Mini-Sampling

```
# Input: data, budget=50 (say), few=64 (say)
# Output: out
out = [rows[0]]
while len(out) < budget:
    tmp = sample(rows, min(few, len(data.rows)))
    w = {r: min(D(data, r, c)**2 for c in out) for r in tmp}
    r1 = random.choices(tmp, weights=[w[r] for r in tmp])[0]
    out.append(r1)
return out
```

Algorithm 2: LITE and EZR — Unified Active Learner

```
# Input: data, start=4 (say), Stop=50 (say), acq in {lite, ezr}
# Output: o(best, rest, todo)
# Note: add, sub, clone, ydists, mostBest, nearer defined in Algs 3,4
if shuffle: random.shuffle(data.rows)
n       = the.start
todo    = data.rows[n:]
br      = clone(data, data.rows[:n])
done    = ydists(data.rows[:n], br)
cut     = round(n**the.guess)
best    = clone(data, done[:cut])
rest    = clone(data, done[cut:])
acquire = mostBest if acq=="lite" else nearer

while len(todo) > 2 and n < the.Stop:
    n  += 1
    hi  = acquire(br, best, rest, todo)
    add(hi, best)
    add(hi, br)
    best.rows = ydists(best.rows, br)
    if len(best.rows) >= round(n**the.guess):
        add(sub(best.rows.pop(-1), best), rest)
return o(best=best, rest=rest, todo=todo)
```

Algorithm 3: MOSTBEST — Acquisition for LITE

```
# Input: xy, best, rest, x, few=64 (say)
# Output: a row removed from x
def score(row):
    return likes(best, row, x.n, 2) / (
           likes(rest, row, x.n, 2) + 1e-32)

*tmp, guess = sorted(x.rows[:few], key=score)
x.rows = tmp + x.rows[few:]
return guess
```

`likes` uses Bayes Theorem to estimate likelihoods. `add` and `sub` update frequency counts inside a Naive Bayes classifier.

Algorithm 4: NEARER — Acquisition for EZR

```
# Input: xy, best, rest, todo, few=64 (say)
# Output: a row removed from todo
bmid = mids(best)
rmid = mids(rest)
for _ in range(the.Few):
    i   = random.randrange(len(todo))
    row = todo[i]
    if distx(xy, bmid, row) < distx(xy, rmid, row):
        return todo.pop(i)
return todo.pop()
```

`mids` returns per-column medians. `distx` computes Euclidean distance over x-attributes. Unlike MOSTBEST, no Naive Bayes model is needed — proximity to `best`'s centroid is the sole criterion.

Mean and Ranks of $d2h$ by Budget

Table 1: Means from optimization trials. For $d2h$ values, smaller is better. Cell colors indicate Scott-Knott rank: best (a), 2nd (b), 3rd (c), 4th (d), 5th (e), 6th (f), 7th (g).

Δ	Rows	[X]	[Y]	As is	LITE										EZR										DEHB										LINE										SMAC										Random									
					6	12	18	24	50	100	200		6	12	18	24	50	100	200		6	12	18	24	50	100	200		6	12	18	24	50	100	200		6	12	18	24	50	100	200		6	12	18	24	50	100	200		6	12	18	24	50	100	200	
95	259	4	2	71	51	48	44	42	30	23	23	49	38	36	36	27	23	23	44	40	36	31	29	25	23	44	40	36	36	32	32	32	34	46	36	31	29	25	23	23	23	23	23	23	23	45	39	36	30	25	23	23	45	39	36	30	25	23	23	SS-H
90	196	3	2	55	43	42	40	38	36	34	33	46	43	42	41	36	32	32	40	40	36	36	32	32	32	41	40	38	35	33	32	32	33	41	39	39	36	35	33	32	44	40	38	37	35	34	34	41	39	39	36	35	33	32	SS-D
90	350	20	1	36	19	15	9	5	1	0	0	12	9	5	4	1	0	0	11	9	6	4	1	0	0	19	6	5	3	1	0	0	0	22	12	7	4	1	0	0	8	7	4	1	0	0	8	7	4	1	0	0	8	7	4	1	0	0	socks
92	192	9	1	33	6	6	5	4	2	1	0	7	3	2	1	0	0	0	9	5	3	2	1	0	0	6	3	2	2	0	0	0	0	8	5	4	2	2	1	0	8	5	4	2	2	1	0	8	5	4	2	2	1	0	8	5	4	2	2	1	Apache_AllMeasurements
93	206	2	1	37	34	30	25	21	10	3	3	32	22	21	15	12	8	3	22	21	15	13	2	0	0	23	21	15	12	8	3	3	3	28	23	21	16	9	4	2	28	23	21	16	9	4	2	28	23	21	16	9	4	2	28	23	21	16	9	4	SS-B
97	247	22	2	84	89	80	80	84	89 h	91 h	78	70	68	66	61	53	46	46	71	68	68	65	63	56	48	74	71	65	60	58	54	49	49	76	66	59	48	47	71	79	79	79	79	79	79	79	79	79	79	79	79	79	79	79	79	79	79	79	79	79	wallpaper
96	196	3	1	27	5	3	2	1	0	0	0	8	2	1	0	0	0	0	8	3	1	1	0	0	0	6	2	1	1	0	0	0	0	3	1	0	0	0	0	0	3	1	0	0	0	0	3	1	0	0	0	0	3	1	0	0	0	0	wc+rs-3d-c4-obj1
97	196	3	1	26	5	2	1	0	0	0	0	8	2	1	0	0	0	0	8	2	1	1	0	0	0	6	2	1	1	0	0	0	0	3	1	0	0	0	0	0	3	1	0	0	0	0	3	1	0	0	0	0	3	1	0	0	0	0	wc+sol-3d-c4-obj1
90	398	4	3	56	38	35	34	34	29	24	19	37	29	26	24	21	18	17	37	29	29	27	23	20	18	37	32	29	29	25	21	18	auto93
87	81	26	1	35	15	14	11	9	6	3	1	11	9	7	6	3	2	1	11	9	7	6	3	2	1	17	14	9	7	5	3	3	player_statistics_cleaned_final
91	196	3	2	57	44	42	42	40	37	34	33	44	42	41	40	38	36	33	44	42	41	40	38	35	33	44	41	40	39	37	35	34	SS-G
96	1152	16	1	27	6	3	2	2	1	0	0	6	1	2	1	0	0	0	10	4	1	0	0	0	0	7	2	2	0	0	0	0	wc+wc-3d-c4-obj1
96	1152	16	1	38	6	4	3	1	1	0	0	4	3	1	1	0	0	0	5	3	2	1	1	0	0	5	2	2	1	1	0	0	X264_AllMeasurements
33	48	38	41	40	43	43	42	37	32	30	48	43	40	42	37	32	30	47	45	41	41	37	35	31	35	48	46	41	42	38	37	33	A2C_Acrobot
17	38	28	20	16	30	21	20	17	17	16	16	31	27	20	18	16	16	44	29	23	21	19	17	17	16	36	27	27	22	19	17	17	SS-P
0	1	0	0	0	0	0	0	0	0	0	0	2	1	0	0	0	0	0	2	1	0	0	0	0	0	1	0	0	0	0	0	0	HSQLDB
43	54	50	47	44	40	39	39	53	52	47	45	41	39	39	47	44	41	41	39	39	47	45	44	44	41	40	39	nasa93dem
2	14	5	3	2	0	0	0	18	10	4	3	2	1	0	0	0	0	0	19	14	8	8	3	2	1	0	0	0	0	0	0	0	LLVM
35	46	44	43	41	35	35	46	43	44	42	39	38	35	35	45	43	41	41	38	37	35	44	41	41	39	38	36	SS-F
17	24	20	19	17	14	13	13	25	20	20	17	16	13	12	25	22	20	18	16	14	13	27	22	20	19	17	15	SS-O
17	38	28	20	16	16	16	16	36	31	25	19	16	16	16	43	34	23	24	18	17	16	36	27	27	22	19	17	SS-L
1	9	5	3	3	1	1	0	8	5	4	4	1	0	0	10	6	5	5	4	3	2	11	6	6	5	4	3	2	SS-A
0	2	0	0	0	0	0	0	4	2	0	0	0	0	0	3	2	1	1	0	0	0	4	3	1	1	0	0	0	0	0	0	0	SS-M
0	25	0	0	0	0	0	0	25	30	5	10	0	0	0	50	15	5	0	0	0	30	0	5	0	0	0	0	student_dropout
18	38	31	23	21	19	18	16	31	32	26	20	17	16	16	31	26	25	21	21	18	17	32	23	25	21	20	18	17	coc1000
22	31	29	28	28	25	22	20	33	28	28	25	23	20	17	31	30	28	28	25	19	18	31	29	29	26	23	22	20	Wine_quality
1	3	1	1	1	1	0	0	3	2	2	1	1	1	0	3	2	2	1	1	1	0	3	2	2	1	1	1	sd-6d-c2-obj1
5	32	20	15	12	8	7	4	33	30	32	17	12	6	3	26	23	18	16	9	6	5	32	25	18	13	12	7	5	SS-K
36	56	52	50	47	41	37	34	51	47	42	41	36	29	28	54	49	45	40	39	34	33	56	48	49	40	40	36	33	Employee-Attrition
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	storm
0	7	3	2	1	0	0	0	7	4	2	1	1	0	0	5	1	1	0	0	0	0	6	2	1	1	0	0	wc-6d-c1-obj1
30	40	38	31	30	29	28	27	39	37	33	31	30	29	39	36	36	34	31	30	40	38	36	35	34	31	30	Scrum1k
17	24	23	20	17	15	13	13	25	22	20	19	17	16	16	21	20	19	19	18	17	15	21	20	19	18	18	17	SS-R
0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	BDBC
0	4	0	0	0	0	0	0	4	2	0	0	0	0	0	4	1	1	0	0	0	0	3	1	1	0	0	0	rs-6d-c3_obj1
4	38	14	12	7	5	4	4	34	29	19	20	5	3	3	21	9	10	8	5	5	4	24	16	13	9	6	6	5	SS-J
5	15	8	7	5	5	4	4	12	7	6	5	5	5	5	14	9	7	6	5	5	5	15	9	8	7	6	5	SS-Q
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	rs-6d-c3_obj2
0	2	1	0	0	0	0	0	2	1	1	0	0	0	0	2	1	1	1	0	0	0	2	1	1	1	1	0	HSMGP_num
0	6	3	2	2	2	1	0	8	5	2	1	1	0	0	5	3	2	1	1	0	0	10	5	3	2	1	0	SS-S
34	43	36	36	36	34	32	29	42	41	41	40	37	33	33	43	40	41	40	38	34	34	43	42	41	40	38	37	35	deparch
11	4	0	0	0	0	0	0	12	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	Health-ClosedIssues0001
0	15	0	0	0	0	0	0	31	6	3	0	3	0	0	21	12	12	3	0	0	0	40	21	3	0	0	0	Health-ClosedPRs0005
20	40	32	28	26	23	21	18	38	33	30	28	24	22	21	40	35	31	30	25	24	22	38	34	30	30	26	23	Health-ClosedPRs0007
16	8	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	Health-ClosedIssues0008
33	56	56	55	55	54	53	51	58 h	54	48	45	40	38	33	52	53	50	51	44	42	34	54	50	50	47	44	36	Health-ClosedPRs0011
45	53	45	44	43	40	37	34	60	50	45	50	48	39	37	50	49	49	44	40	37	34	54	50	50	49	50	47	home_data_for_ml_course
23	40	31	28	26	24	21	18	41	37	36	32	28	24	20	37	35	31	32	26	24	21	38	37	34	32	29	26	Health-Commits0002
18	31	28	26	24	22	19	14	30	29	27	28	25	22	20	30	27	26	27	24	21	20	30	27	27	26	22	20	Health-Commits0008
2	10	3	3	3	2	2	1	14	6	3	3	3	3	3	2	2	2	2	2	2	2	10	2	2	2	2	2	Health-Commits0006
28	60	44	43	42	40	39	38	59	41	35	32	28	28	28	57	57	46	34	28	28	28	66	53	43	38	28	28	Health-ClosedPRs0002
0	20	2	0	0	0	0	0	25	10	5	10	0	0	0	25	18	2	0	0	0	0	35	8	5	0	0	0	Health-ClosedIssues0004
65	79	67	64	62	58	57	57	80	72	67	66	62	57	56	79	69	68	63	58	57	56	81	80	71	65	59	57	dataset600
16	9	0	0	0	0	0	0	16	0	0	0	0	0	0	7	0	0	0	0	0	0	19	11	0	0	0	0	Health-ClosedIssues0011
6	17	14	13	10	6	3	2	18	12	10	7	4	2	2	20	15	13	11	9	6	4	15	10	7	5	2	0	Health-ClosedPRs0008
26	54	45	41	40	34	27	20	51	52	49	47	38	29	22	55 h	49	47	46	40	37	36	54 h	52	48	47	46	40	Life_Expectancy_Data
23	30	29	28	26	26	24	21	30	29	29	29	26	24	22	29	28	27	27	25	24	24	30	28	27	27	26	25	Health-ClosedPRs0010
21	37	31	29	28	26	24	21	37	29	29	30	29	26	24	31	29	30	29	28	25	23	32	31	29	29	27	25	Health-Commits0004
4	10	6	6	5	5	5	5	11	7	6	6	4	4	3	14	12	12	11	10	9	9	13	12	11	11	10	9	Health-Commits0010
19	39	35	33	32	29	27	24	39	38	34	30	28	26	22	43	34	33	34	27	22	20	41	34	34	31	27	17	Health-Commits0009
14	23	17	17	17	15	14	11	33	18	18	17	16	15	14	17	18	18	17	16	15	15	24	18	18	18	17	17	SS-T
54 i	46 h	31	31	25	14	11	8	52 h	40	37	34	11	5	2	45	42	34	28	10	8	2	57 i	40	40	36	22	12	redis
40	45	35	31	29	23	19	15	43	37	35	31	27	23	17	38	38	35	34	27	24	21	43	39	38	32	30	22	Health-ClosedIssues0002
16	20	16	16	16	16	16	16	27	16	16	16	16	16	16	16	16	16	16	16	16	16	23	16	16	16	16	16	Health-ClosedPRs0006
22	27	23	21	19	19	15	13	22	20	18	18	14	13	12	28	22	20	20	18	14	14	28	24	23	18	18	13	Health-Commits0003
26	45 h	39	33	32	29	19	15	42	39	34	37	27	20	15	44 h	34	36	35	27	24	20	42	40	34	34	30	24	Health-Commits0001
23	22	22	22	22	22	22	21	23	23	23	23	21	20	15	23	23	22	22	22	22	21	23	23	22	22	22	21	Health-ClosedIssues0003
2	13	8	6	6	4	4	3	12	8	7	7	4	3	2	10	10	6	7	5	3	3	10	9	7	6	5	4	Health-ClosedIssues0009
49	49	46	45	45	45	45	42	49	47	47	46	46	45	43	48	48	47	47	45	45	41	49	48	47	47	46	45	Health-ClosedIssues0004
8	8	8	8	8	8	8	8	23	8	8	8	8	8	8	8	8	8	8	8	8	8	15	8	8	8	8	8	Health-ClosedIssues0006
32 h	60	54	54	49	30	26	23	62 h	59	51	51	53	38	33	58	60	57	48	40	38	21	60	55	57	52	47	43	Health-ClosedIssues0010
2	26	23	22	22	21	21	20	28	24	23	23	22	21	21	25	24	24	23	22	22	21	25	24	23	23	22	21	Health-ClosedIssues0007
2	26	14	12	9	4	1	1	31	21	18	11	4	2	2	22	18	12	11	9	4	3	31	27	17	11	9	4	SS-V
52	65	61	60	58	54	54	52	65	62	59	60	55	51	49	64	59	59	57	56	53	51	65	62	57	57	54	53	Health-ClosedIssues0005
19	34	22	19	19	17	13	10	34	26	26	20	17	13	10	34	30	25	25	20	18	15	38	31	26	25	23	19	Health-Commits0000
25	34	30	28	25	24	23	23	33	26	26	25	24	24	24	27	25	25	24	24	24	23	27	25	25	24	24	23	SS-U
2	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Health-ClosedPRs0000
9	30	14	10	10	9	8	8	51	32	21	18	10	8	8	31	21	17	16	12	10	8	32	29	18	18	12	8	Health-Commits0007
66	10	0	0	0	0	0	0	13	3	0	0	0	0	0	3	3	0	0	0	0	0	13	3	0	0	0	0	Health-Commits0011
4	8	2	2	1	1	0	0	4	3	1	1	0	0	0	1	8	6	2	1	1	0	9	3	2	2	1	0	Telco-Customer-Churn
1	1	0	0	0	0	0	0	2	1	0	0	0	0	0	1	1	1	1	0	0	0	1	1	1	1	0	0	PostgreSQL
54	61	54	51	50	42	34	27	62	59	60	52	49	41	29	63	54	58	51	49	39	36	67	60	56	57	54	47	Health-ClosedPRs0009
1	5	4	3	2	1	1	0	6	3	3	2	1	1	1	6	3	3	2	1	1	1	5	3	2	1	1	1	dconvert
1	35	29	26	22	17	14	10	34	29	24	24	21	20	17	39	33	30	28	24	19	18	39	32	30	28	23	19	SS-X
2	36	30	26	25	22	19	14	35	31	28	22	20	17	17	35	31	28	26	21	21	22	34	30	30	29	26	23	pom3c
14	26	17	15	14	11	9	7	27	21	13	15	9	7	3	25	19	17	16	12	9	8	22	20	16	15	14	11	SQL_AllMeasurements
27	41	37	32	31	25	22	18	40	37	32	30	28	25	23	39	36	33	32	27	27	23	38	33	32	31	29	24	SS-W
1	6	5	4	3	2	2	1	8	5	4	4	2	2	1	8	5	4	3	3	2	1	7	4	4	4	2	2	exastencils
7	21	16	13	10	8	5	4	20	13	11	8	6	4	2	19	14	15	13	11	7	6	20	16	15	13	11	8	pom3b
6	19	13	10	8	6	5	4	21	13	10	10	8	6	4	19	16	14	11	10	8	7	20	17	13	12	10	8	pom3a
0	1	1	1	1	0	0	0	1	1	1	0	0	0	0	1	1	1	0	0	0	0	1	1	1	0	0	0	7z
38	55	46	43	41	39	36	33	61	54	45	43	40	38	37	57	49	49	46	44	40	38	55	53	50	47	43	41	BankChurners
12	26	22	22	21	21	21	21	27	21	18	14	12	11	10	25	21	20	16	14	12	10	25	23	22	19	18	15	x264
10	20	17	16	15	11	9	8	26	19	16	17	12	10	8	19	16	15	13	11	9	8	19	17	14	13	11	10	SS-N
12	26	19	15	15	11	9	8	26	19	16	17	12	10	8	23	18	19	16	15	14	12	22	20	18	18	14	11	xom.osp2
4	14	10	8	7	4	2	2	14	8	7	5	3	2	1	15	11	9	8	7	5	4	12	9	7	6	4	2	xom.osp
4	13	8	6	6	4	2	2	13	8	7	5	3	2	1	14	11	7	6	4	3	1	13	9	9	7	6	5	xom.flight
48	78	75	72	70	66	64	62	75	63	57	59	44	40	37	78	74	63	64	60	53	51	71	67	63	70	53	52	Data_COVID19_Indonesia
36	44	42	40	39	37	35	35	44	43	40	40	39	38	37	42	43	41	40	39	38	37	44	43	40	41	39	38	billing10k
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	javage
31	40	36	34	33	30	29	28	43	40	37	36	33	32	31	44	39	37	36	35	34	32	40	40	38	36	36	33	Scrum10k
33	41	34	31	30	27	26	25	41	36	35	34	30	29	28	40	37	36	36	33	29	29	42	37	35	35	33	31	FFM-125-25-0.50-SAT-1
34	45	41	39	39	35	34	33	45	43	40	39	38	36	35	43	41	40	39	38	36	35	43	41	40	39	38	37	Scrum100k
37	45	40	38	37	35	34	33	44	42	41	39	37	37	36	43	42	40	39	38	36	35	42	42	42	40	38	37	FFM-250-50-0.50-SAT-1
30	43	36	34	33	30	29	28	44	42	40	39	37	36	35	43	42	40	40	37	36	35	42	42	42	40	38	37	FFM-500-100-0.50-SAT-1
31	43	36	34	33	30	29	28	44	42	40	39	37	36	35	43	42	40	40	37	36	35	42	42	42	40	38	37	FFM-500-100-1.00-SAT-1
30	43	36	34	33	30	29	28	44	42	40	39	37	36	35	43	42	40	40	37	36	35	42	42	42	40	38	37	FFM-500-100-0.75-SAT-1
36	45	39	38	37	35	34	34	45	42	39	39	37	36	36	43	41	40	40	38	37	36	43	43	40	40	37	37	FFM-500-100-0.25-SAT-1
30	44	36	33	33	29	27	27	43	39	36	35	32	31	30	44	41	39	37	35	33	30	42	39	36	37	35	32	FFM-500-100-0.50-SAT-1
83	32	54	70	75	82	88	94	34	47	56	62	82	90	94	36	43	54	59	78	87	91	35	43	48	57	75	85	91
				%Best	0										24	25	31	39	59	64	69	28	38	49	63	79	86	90	27	41	51	55	79	83	83	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32

Standard Deviations of $d2h$ by Budget

Seven tables follow, one per budget (6, 12, 18, 24, 50, 100, 200). Each table shows all 7 optimizers side-by-side for all datasets, enabling direct cross-optimizer comparison at a fixed budget. Smaller SD values indicate more stable (repeatable) results.

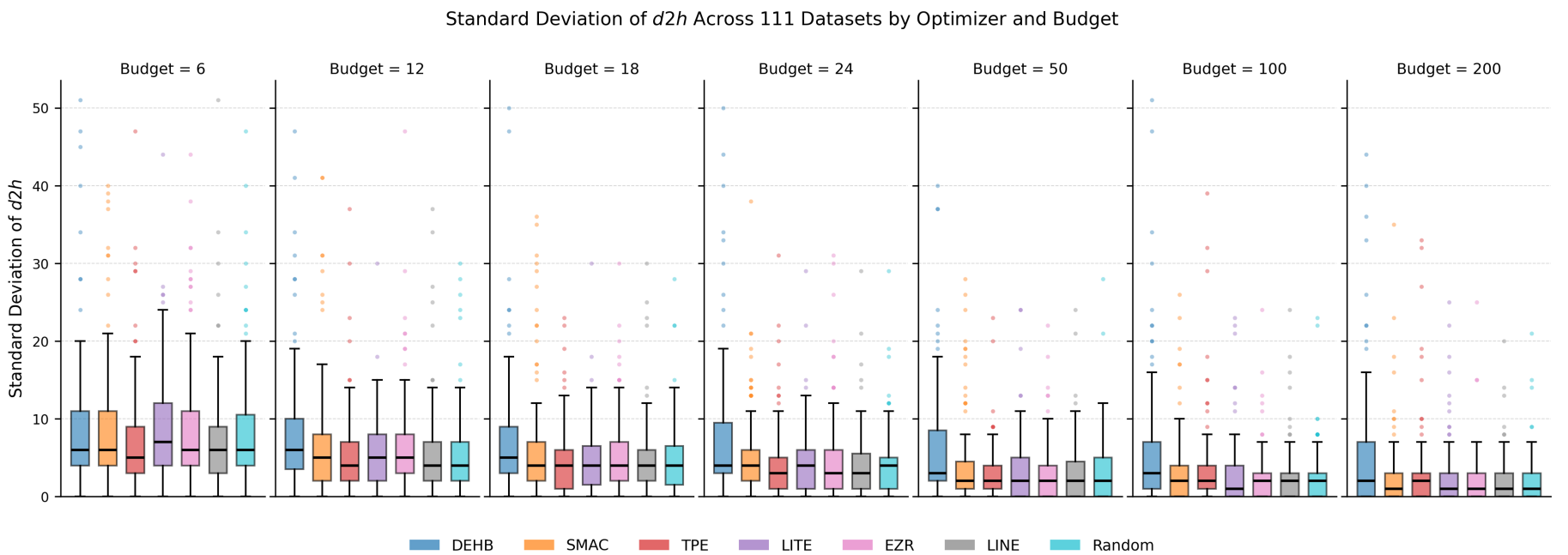

Figure 1: Standard deviation of $d2h$ across 111 datasets by optimizer and budget. Box plots show the distribution over datasets; lower values indicate more stable (reproducible) results.

Budget = 6 evaluations

SD of $d2h$ at budget 6 — all optimizers vs. all datasets

ID	Δ	Rows	$\|X\|$	$\|Y\|$	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
1	95	259	4	2	16	12	13	11	9	16	15	SS-H
2	90	196	3	2	4	6	8	5	4	6	6	SS-D
3	90	350	20	1	15	11	12	7	8	5	11	socks
4	92	192	9	1	4	5	5	5	5	6	5	Apache_AllMeasurements
5	93	206	3	2	13	12	11	13	12	10	13	SS-B
6	97	247	22	2	11	8	8	4	11	8	4	wallpaper
7	96	196	3	1	8	9	6	11	8	8	8	wc+rs-3d-c4-obj1
8	97	196	3	1	9	9	6	11	6	9	8	wc+sol-3d-c4-obj1
9	90	398	4	3	10	8	5	8	8	8	7	auto93
10	87	81	26	1	8	11	6	8	9	6	10	player_statistics_cleaned_final
11	91	196	3	2	5	6	8	5	5	4	7	SS-G
12	96	196	3	1	9	9	6	11	9	10	9	wc+wc-3d-c4-obj1
13	96	1152	16	1	6	2	3	4	3	5	6	X264_AllMeasurements
14	84	223	10	3	4	7	6	6	7	8	6	A2C_Acrobot
15	95	1023	11	2	17	13	18	14	17	17	15	SS-P
16	98	864	18	1	1	2	2	1	2	2	2	HSQLDB
17	95	93	22	3	5	6	0	3	3	5	4	nasa93dem
18	92	1024	10	1	11	17	13	12	13	13	16	LLVM
19	92	196	3	2	4	6	7	5	4	3	6	SS-F
20	82	972	11	2	6	6	4	7	6	6	6	SS-O
21	95	1023	11	2	17	13	18	16	17	13	15	SS-L
22	83	1343	3	2	4	5	4	4	3	4	6	SS-A
23	97	864	17	3	3	4	4	3	4	4	5	SS-M
24	97	649	33	1	47	37	47	44	44	51	47	student_dropout
25	81	1000	20	5	7	9	5	6	10	8	8	coc1000
26	79	1599	10	2	6	4	5	6	6	6	3	Wine_quality
27	93	2866	6	1	6	6	3	2	1	5	2	sol-6d-c2-obj1
28	89	2880	6	2	11	12	12	12	13	13	12	SS-K
29	78	1470	32	3	9	11	9	11	9	13	13	WA_Fn-UseC_-HR-Employee-Attrition
30	100	1557	12	1	0	0	0	1	2	1	2	storm
31	94	2880	6	1	10	8	6	8	5	6	7	wc-6d-c1-obj1
32	86	1000	124	3	5	3	5	5	5	5	5	Scrum1k
33	76	3008	14	2	6	3	4	7	8	5	5	SS-R
34	99	2560	16	1	3	1	0	0	1	0	1	BDBC
35	97	3840	6	1	6	5	2	5	5	5	4	rs-6d-c3_obj1
36	91	3840	6	2	16	17	18	18	16	15	20	SS-J

Continued...

Budget 6 — continued

ID	Δ	Rows	|X|	|Y|	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
37	92	2736	13	3	11	8	13	13	8	8	10	SS-Q
38	100	3840	6	1	0	0	0	0	0	0	0	rs-6d-c3_obj2
39	95	3456	14	1	1	1	1	1	1	2	1	HSMGP_num
40	92	3840	6	2	5	4	2	4	3	3	5	SS-S
41	92	4096	12	1	7	7	8	8	7	7	9	deeparch
42	49	10000	5	3	12	2	8	2	3	3	3	Health-ClosedIssues0001
43	98	10000	5	3	40	38	29	18	29	0	24	Health-ClosedPRs0005
44	96	10000	5	3	14	31	29	27	32	30	30	Health-ClosedPRs0007
45	81	10000	5	3	11	9	9	10	11	9	9	Health-ClosedIssues0008
46	98	10000	5	3	0	40	0	24	24	0	24	Health-ClosedPRs0011
47	95	1460	77	4	4	3	9	7	4	9	4	home_data_for_ml_course
48	47	10000	5	3	12	19	1	13	17	0	12	Health-Commits0002
49	81	10000	5	3	8	4	7	8	7	7	6	Health-Commits0008
50	60	10000	5	3	17	4	3	3	3	4	3	Health-Commits0006
51	97	10000	5	3	28	28	0	23	28	0	24	Health-ClosedPRs0002
52	93	10000	5	3	4	26	20	22	24	22	20	Health-ClosedPRs0004
53	97	5160	9	1	34	21	22	25	38	34	40	dataset600
54	76	10000	5	3	0	12	10	12	12	13	10	Health-ClosedPRs0008
55	97	10000	5	3	45	39	1	26	32	22	34	Health-ClosedIssues0011
56	84	2938	20	2	9	7	5	5	6	7	6	Life_Expectancy_Data
57	81	10000	5	3	7	5	2	6	11	5	6	Health-ClosedPRs0010
58	43	10000	5	3	3	3	2	2	2	2	2	Health-ClosedIssues0004
59	67	10000	5	3	12	4	3	12	12	3	4	Health-Commits0010
60	77	10000	5	3	3	3	5	2	2	2	2	Health-Commits0009
61	91	5184	12	2	2	2	2	7	3	3	2	SS-T
62	60	4400	9	1	8	6	11	10	8	6	8	redis
63	80	10000	5	3	14	1	30	19	27	3	19	Health-ClosedIssues0002
64	98	10000	5	3	0	13	13	26	21	26	13	Health-ClosedPRs0006
65	88	10000	5	3	12	5	4	6	10	7	5	Health-Commits0003
66	95	10000	5	3	28	32	1	15	25	0	21	Health-Commits0001
67	75	10000	5	3	10	7	8	6	6	6	5	Health-ClosedIssues0003
68	79	10000	5	3	9	8	2	2	8	2	6	Health-ClosedIssues0009
69	28	10000	5	3	1	1	1	3	1	0	0	Health-Commits0004
70	81	10000	5	3	8	6	6	7	6	6	6	Health-ClosedIssues0006
71	33	10000	5	3	1	4	1	1	1	3	2	Health-ClosedIssues0010
72	97	10000	5	3	1	0	0	22	28	0	22	Health-ClosedPRs0003
73	72	10000	5	3	11	1	0	11	2	11	1	Health-ClosedIssues0007
74	90	6840	16	2	14	12	11	12	10	10	7	SS-V
75	61	10000	5	3	24	3	14	11	7	1	3	Health-ClosedIssues0005
76	81	2205	31	8	2	4	4	6	6	6	5	Marketing_Analytics
77	74	10000	5	3	20	21	20	15	15	12	17	Health-Commits0000
78	89	4608	21	2	5	6	7	9	10	4	3	SS-U
79	98	10000	5	2	51	22	0	0	0	0	0	Health-ClosedPRs0000
80	91	10000	5	3	16	20	17	24	17	18	19	Health-Commits0007
81	97	10000	5	3	0	31	32	24	27	14	27	Health-Commits0011
82	92	7043	19	2	11	10	5	9	5	8	11	WA_Fn-UseC_-Telco-Customer-Churn
83	95	19008	9	1	0	0	1	0	2	0	0	PostgreSQL
84	71	10000	5	3	8	8	6	14	13	9	6	Health-ClosedPRs0009
85	89	6764	18	1	4	5	3	4	4	4	4	dconvert
86	96	86058	11	2	15	16	15	8	10	9	8	SS-X
87	82	20000	9	3	5	4	5	4	8	5	8	pom3c
88	70	4653	38	1	7	11	9	9	8	10	9	SQL_AllMeasurements
89	82	65536	16	2	7	6	5	5	5	4	5	SS-W
90	82	86058	12	1	5	5	3	3	4	3	4	exastencils
91	78	20000	9	3	6	8	7	8	7	7	7	pom3b
92	80	20000	9	3	7	9	8	7	7	6	6	pom3a
93	95	68640	14	1	1	2	1	1	1	1	1	7z
94	72	10127	19	4	18	14	11	13	14	9	12	BankChurners
95	62	53662	17	1	5	6	6	5	4	3	3	x264
96	58	53662	17	2	5	5	4	5	3	2	1	SS-N
97	80	10000	24	4	5	5	4	9	9	6	5	xomo_osp2
98	81	10000	24	4	4	4	4	6	4	6	5	xomo_osp
99	81	10000	24	4	6	5	4	4	6	6	5	xomo_flight
100	92	21759	35	3	9	12	10	13	14	12	18	Data_COVID19_Indonesia
101	79	10000	88	3	3	3	4	4	4	3	4	billing10k
102	93	166975	35	1	0	0	0	0	0	0	0	javagc
103	84	10000	124	3	4	4	5	6	5	6	5	Scrum10k
104	86	10000	128	3	7	5	4	6	6	4	6	FFM-125-25-0.50-SAT-1
105	81	100000	124	3	3	4	4	3	4	5	3	Scrum100k
106	85	10000	256	3	3	4	4	5	4	3	3	FFM-250-50-0.50-SAT-1
107	83	10000	510	3	4	6	4	7	6	5	4	FFM-500-100-0.50-SAT-1
108	85	10000	511	3	6	5	5	5	6	4	5	FM-500-100-1.00-SAT-1
109	85	10000	511	3	5	4	5	6	6	6	4	FM-500-100-0.75-SAT-1
110	81	10000	517	3	3	3	3	5	4	3	4	FM-500-100-0.25-SAT-1
111	82	10000	513	3	5	4	6	7	6	5	6	FM-500-100-0.50-SAT-1

Budget = 12 evaluations

SD of $d2h$ at budget 12 — all optimizers vs. all datasets

ID	Δ	Rows	|X|	|Y|	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
1	95	259	4	2	15	11	10	6	12	13	11	SS-H
2	90	196	3	2	4	5	5	1	6	5	4	SS-D
3	90	350	20	1	13	9	5	6	6	6	7	socks
4	92	192	9	1	3	4	4	2	3	4	3	Apache_AllMeasurements
5	93	206	3	2	11	12	9	15	10	11	13	SS-B
6	97	247	22	2	6	8	12	9	13	12	7	wallpaper
7	96	196	3	1	6	4	2	1	2	4	2	wc+rs-3d-c4-obj1
8	97	196	3	1	5	2	2	0	4	3	2	wc+sol-3d-c4-obj1
9	90	398	4	3	8	8	5	8	6	8	7	auto93
10	87	81	26	1	8	7	6	5	4	6	6	player_statistics_cleaned_final
11	91	196	3	2	4	6	5	3	5	4	5	SS-G
12	96	196	3	1	6	2	2	1	2	6	2	wc+wc-3d-c4-obj1
13	96	1152	16	1	7	2	2	1	5	3	2	X264_AllMeasurements
14	84	223	10	3	5	6	7	2	8	3	4	A2C_Acrobot
15	95	1023	11	2	16	12	13	13	12	11	12	SS-P
16	98	864	18	1	0	0	0	1	1	2	2	HSQLDB
17	95	93	22	3	6	3	2	5	5	3	5	nasa93dem
18	92	1024	10	1	9	11	8	5	9	8	11	LLVM
19	92	196	3	2	5	5	5	4	5	4	4	SS-F
20	82	972	11	2	5	5	5	6	4	5	5	SS-O
21	95	1023	11	2	16	12	13	12	15	12	12	SS-L
22	83	1343	3	2	17	3	4	3	3	3	3	SS-A
23	97	864	17	3	2	3	0	2	3	3	3	SS-M
24	97	649	33	1	41	0	0	0	47	37	0	student_dropout
25	81	1000	20	5	5	6	6	9	9	7	5	coc1000
26	79	1599	10	2	6	5	4	3	6	4	5	Wine_quality
27	93	2866	6	1	1	1	0	0	1	0	1	sol-6d-c2-obj1
28	89	2880	6	2	12	12	11	9	14	11	13	SS-K
29	78	1470	32	3	10	7	8	10	9	9	10	WA_Fn-UseC_-HR-Employee-Attrition
30	100	1557	12	1	0	0	0	0	0	0	0	storm
31	94	2880	6	1	8	3	3	5	5	2	3	wc-6d-c1-obj1
32	86	1000	124	3	5	5	5	4	5	5	4	Scrum1k
33	76	3008	14	2	6	5	3	8	8	2	3	SS-R
34	99	2560	16	1	0	0	0	0	0	0	0	BDBC
35	97	3840	6	1	5	2	1	0	4	1	2	rs-6d-c3_obj1
36	91	3840	6	2	16	14	11	8	19	4	10	SS-J
37	92	2736	13	3	9	5	5	8	3	4	4	SS-Q
38	100	3840	6	1	0	0	0	0	0	0	0	rs-6d-c3_obj2
39	95	3456	14	1	1	1	1	1	1	1	1	HSMGP_num
40	92	3840	6	2	3	2	2	2	3	2	2	SS-S
41	92	4096	12	1	7	5	5	5	5	5	6	deeparch
42	49	10000	5	3	16	6	3	8	3	4	2	Health-ClosedIssues0001
43	98	10000	5	3	0	0	0	0	0	0	0	Health-ClosedPRs0005
44	96	10000	5	3	1	31	14	0	19	25	30	Health-ClosedPRs0007
45	81	10000	5	3	8	8	5	8	9	6	8	Health-ClosedIssues0008
46	98	10000	5	3	0	41	37	0	0	0	0	Health-ClosedPRs0011
47	95	1460	77	4	14	5	5	7	7	4	7	home_data_for_ml_course
48	47	10000	5	3	12	0	4	10	0	7	0	Health-Commits0002
49	81	10000	5	3	10	4	6	6	6	6	4	Health-Commits0008
50	60	10000	5	3	10	4	3	5	4	4	5	Health-Commits0006
51	97	10000	5	3	34	0	0	1	17	0	0	Health-ClosedPRs0002
52	93	10000	5	3	2	25	23	13	21	22	23	Health-ClosedPRs0004
53	97	5160	9	1	47	26	15	11	21	34	24	dataset600
54	76	10000	5	3	7	6	14	13	14	14	11	Health-ClosedPRs0008
55	97	10000	5	3	20	41	0	0	0	0	26	Health-ClosedIssues0011
56	84	2938	20	2	4	8	8	6	6	5	7	Life_Expectancy_Data
57	81	10000	5	3	1	10	8	8	6	6	5	Health-ClosedPRs0010
58	43	10000	5	3	3	2	3	2	3	1	2	Health-ClosedIssues0004
59	67	10000	5	3	12	4	2	8	3	4	2	Health-Commits0010
60	77	10000	5	3	9	2	2	2	2	2	2	Health-Commits0009
61	91	5184	12	2	2	2	2	2	2	2	2	SS-T
62	60	4400	9	1	8	8	9	7	7	10	7	redis
63	80	10000	5	3	19	29	15	4	3	2	0	Health-ClosedIssues0002
64	98	10000	5	3	28	24	30	30	29	27	28	Health-ClosedPRs0006
65	88	10000	5	3	9	6	4	11	9	8	7	Health-Commits0003
66	95	10000	5	3	15	0	1	0	0	0	0	Health-Commits0001
67	75	10000	5	3	7	7	7	5	6	6	6	Health-ClosedIssues0003
68	79	10000	5	3	5	10	8	7	7	10	6	Health-ClosedIssues0009
69	28	10000	5	3	0	0	0	3	0	0	1	Health-Commits0004
70	81	10000	5	3	7	2	6	6	5	6	5	Health-ClosedIssues0006
71	33	10000	5	3	2	4	2	3	2	2	1	Health-ClosedIssues0010
72	97	10000	5	3	26	31	0	0	0	0	0	Health-ClosedPRs0003
73	72	10000	5	3	1	1	1	18	11	0	15	Health-ClosedIssues0007
74	90	6840	16	2	12	6	8	15	10	9	10	SS-V
75	61	10000	5	3	21	14	1	3	1	1	1	Health-ClosedIssues0005

Continued. . .

Budget 12 — continued

ID	Δ	Rows	|X|	|Y|	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
76	81	2205	31	8	3	3	3	6	5	5	5	Marketing_Analytics
77	74	10000	5	3	18	6	6	6	5	11	9	Health-Commits0000
78	89	4608	21	2	5	3	4	9	1	1	1	SS-U
79	98	10000	5	2	31	0	0	0	0	0	0	Health-ClosedPRs0000
80	91	10000	5	3	16	17	14	12	23	15	17	Health-Commits0007
81	97	10000	5	3	28	15	20	0	15	14	14	Health-Commits0011
82	92	7043	19	2	16	14	4	2	4	8	3	WA_Fn-UseC_-Telco-Customer-Churn
83	95	19008	9	1	0	0	0	0	0	0	0	PostgreSQL
84	71	10000	5	3	9	14	5	14	8	15	9	Health-ClosedPRs0009
85	89	6764	18	1	4	2	2	3	2	2	2	dconvert
86	96	86058	11	2	10	9	9	9	8	8	10	SS-X
87	82	20000	9	3	6	4	5	6	5	4	5	pom3c
88	70	4653	38	1	6	7	9	7	7	4	6	SQL_AllMeasurements
89	82	65536	16	2	6	7	4	5	5	5	5	SS-W
90	82	86058	12	1	5	2	2	2	2	3	1	exastencils
91	78	20000	9	3	6	7	5	7	4	6	6	pom3b
92	80	20000	9	3	7	6	6	5	4	5	6	pom3a
93	95	68640	14	1	1	1	1	0	1	1	1	7z
94	72	10127	19	4	9	12	9	9	11	10	10	BankChurners
95	62	53662	17	1	5	4	4	5	5	5	4	x264
96	58	53662	17	2	3	3	2	4	5	3	3	SS-N
97	80	10000	24	4	5	4	4	6	6	6	5	xomo_osp2
98	81	10000	24	4	3	4	3	5	3	5	4	xomo_osp
99	81	10000	24	4	6	4	4	4	4	3	4	xomo_flight
100	92	21759	35	3	9	11	11	15	15	11	13	Data_COVID19_Indonesia
101	79	10000	88	3	3	2	3	3	3	3	3	billing10k
102	93	166975	35	1	0	0	0	0	0	0	0	javagc
103	84	10000	124	3	4	4	3	4	4	3	5	Scrum10k
104	86	10000	128	3	6	5	4	5	3	5	6	FFM-125-25-0.50-SAT-1
105	81	100000	124	3	3	4	3	4	4	3	4	Scrum100k
106	85	10000	256	3	4	3	3	3	4	2	3	FFM-250-50-0.50-SAT-1
107	83	10000	510	3	4	4	4	4	4	5	3	FFM-500-100-0.50-SAT-1
108	85	10000	511	3	6	3	4	4	4	4	5	FM-500-100-1.00-SAT-1
109	85	10000	511	3	5	4	5	6	4	4	5	FM-500-100-0.75-SAT-1
110	81	10000	517	3	3	3	2	3	3	2	3	FM-500-100-0.25-SAT-1
111	82	10000	513	3	6	4	4	5	4	4	5	FM-500-100-0.50-SAT-1

Budget = 18 evaluations

SD of $d2h$ at budget 18 — all optimizers vs. all datasets

ID	Δ	Rows	|X|	|Y|	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
1	95	259	4	2	13	7	10	5	10	11	10	SS-H
2	90	196	3	2	4	5	4	3	4	4	5	SS-D
3	90	350	20	1	11	7	6	5	5	6	7	socks
4	92	192	9	1	3	2	4	1	2	2	3	Apache_AllMeasurements
5	93	206	3	2	10	9	10	10	8	8	10	SS-B
6	97	247	22	2	9	8	13	13	11	9	11	wallpaper
7	96	196	3	1	4	0	0	0	6	2	2	wc+rs-3d-c4-obj1
8	97	196	3	1	4	1	0	0	2	2	1	wc+sol-3d-c4-obj1
9	90	398	4	3	8	7	5	6	6	8	7	auto93
10	87	81	26	1	9	6	7	4	3	6	5	player_statistics_cleaned_final
11	91	196	3	2	4	4	4	3	5	4	4	SS-G
12	96	196	3	1	5	2	0	0	5	2	4	wc+wc-3d-c4-obj1
13	96	1152	16	1	5	2	2	0	2	2	2	X264_AllMeasurements
14	84	223	10	3	6	6	6	3	7	5	4	A2C_Acrobot
15	95	1023	11	2	13	6	12	8	8	9	10	SS-P
16	98	864	18	1	0	1	0	1	0	1	0	HSQLDB
17	95	93	22	3	3	2	6	5	5	2	2	nasa93dem
18	92	1024	10	1	10	8	9	4	4	7	8	LLVM
19	92	196	3	2	5	4	4	3	5	4	4	SS-F
20	82	972	11	2	5	5	5	6	5	5	5	SS-O
21	95	1023	11	2	13	6	12	7	15	8	10	SS-L
22	83	1343	3	2	13	2	2	2	2	2	2	SS-A
23	97	864	17	3	1	2	0	2	0	3	0	SS-M
24	97	649	33	1	50	0	22	0	22	22	22	student_dropout
25	81	1000	20	5	4	3	6	6	7	6	6	coc1000
26	79	1599	10	2	6	5	2	3	4	3	3	Wine_quality
27	93	2866	6	1	1	0	0	0	1	0	0	sol-6d-c2-obj1
28	89	2880	6	2	12	9	9	8	12	8	9	SS-K
29	78	1470	32	3	4	7	7	9	9	8	10	WA_Fn-UseC_-HR-Employee-Attrition
30	100	1557	12	1	0	0	0	0	0	0	0	storm
31	94	2880	6	1	3	1	1	3	2	1	2	wc-6d-c1-obj1
32	86	1000	124	3	4	4	5	3	3	4	4	Scrum1k
33	76	3008	14	2	2	2	3	5	3	3	2	SS-R
34	99	2560	16	1	0	0	0	0	0	0	0	BDBC
35	97	3840	6	1	4	1	0	0	0	1	1	rs-6d-c3_obj1
36	91	3840	6	2	16	4	10	8	15	6	9	SS-J
37	92	2736	13	3	4	5	4	8	2	3	4	SS-Q
38	100	3840	6	1	0	0	0	0	0	0	0	rs-6d-c3_obj2
39	95	3456	14	1	1	0	0	0	0	1	1	HSMGP_num
40	92	3840	6	2	2	1	1	2	2	1	1	SS-S
41	92	4096	12	1	3	5	5	3	3	4	4	deeparch
42	49	10000	5	3	8	2	5	7	2	3	3	Health-ClosedIssues0001
43	98	10000	5	3	0	35	0	0	0	0	0	Health-ClosedPRs0005
44	96	10000	5	3	0	31	14	0	14	25	14	Health-ClosedPRs0007
45	81	10000	5	3	9	8	5	7	8	7	6	Health-ClosedIssues0008
46	98	10000	5	3	18	29	0	0	0	0	0	Health-ClosedPRs0011
47	95	1460	77	4	12	6	10	7	8	6	6	home_data_for_ml_course
48	47	10000	5	3	2	8	16	10	14	6	0	Health-Commits0002
49	81	10000	5	3	6	6	6	7	5	7	6	Health-Commits0008
50	60	10000	5	3	5	5	4	5	6	6	5	Health-Commits0006
51	97	10000	5	3	0	36	15	1	2	0	0	Health-ClosedPRs0002
52	93	10000	5	3	24	24	23	13	18	23	22	Health-ClosedPRs0004
53	97	5160	9	1	47	22	0	0	15	11	15	dataset600
54	76	10000	5	3	9	15	8	12	14	14	14	Health-ClosedPRs0008
55	97	10000	5	3	24	22	19	0	0	0	0	Health-ClosedIssues0011
56	84	2938	20	2	6	4	9	5	4	4	4	Life_Expectancy_Data
57	81	10000	5	3	7	10	6	9	10	7	10	Health-ClosedPRs0010
58	43	10000	5	3	3	5	2	2	3	2	1	Health-ClosedIssues0004
59	67	10000	5	3	12	6	4	2	3	2	3	Health-Commits0010
60	77	10000	5	3	2	1	1	1	1	2	2	Health-Commits0009
61	91	5184	12	2	2	2	2	2	2	1	1	SS-T
62	60	4400	9	1	6	11	8	9	7	6	7	redis
63	80	10000	5	3	14	3	1	4	3	1	0	Health-ClosedIssues0002
64	98	10000	5	3	28	27	0	30	30	30	28	Health-ClosedPRs0006
65	88	10000	5	3	4	5	3	11	10	9	6	Health-Commits0003
66	95	10000	5	3	2	0	1	0	0	0	0	Health-Commits0001
67	75	10000	5	3	6	6	6	5	6	4	6	Health-ClosedIssues0003
68	79	10000	5	3	3	9	8	10	10	8	9	Health-ClosedIssues0009
69	28	10000	5	3	0	0	0	3	0	0	0	Health-Commits0004
70	81	10000	5	3	13	2	2	4	2	4	3	Health-ClosedIssues0006
71	33	10000	5	3	6	4	1	3	2	2	2	Health-ClosedIssues0010
72	97	10000	5	3	1	16	0	0	0	0	0	Health-ClosedPRs0003
73	72	10000	5	3	11	11	11	18	20	11	11	Health-ClosedIssues0007
74	90	6840	16	2	9	6	6	14	11	9	10	SS-V
75	61	10000	5	3	21	1	1	3	1	1	1	Health-ClosedIssues0005

Continued...

Budget 18 — continued

ID	Δ	Rows	$\|X\|$	$\|Y\|$	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
76	81	2205	31	8	3	2	3	5	4	4	6	Marketing_Analytics
77	74	10000	5	3	11	6	5	4	7	5	5	Health-Commits0000
78	89	4608	21	2	3	3	4	7	4	1	1	SS-U
79	98	10000	5	2	22	0	0	0	0	0	0	Health-ClosedPRs0000
80	91	10000	5	3	16	17	3	5	17	8	12	Health-Commits0007
81	97	10000	5	3	0	30	1	0	0	0	0	Health-Commits0011
82	92	7043	19	2	12	20	1	1	1	1	2	WA_Fn-UseC_-Telco-Customer-Churn
83	95	19008	9	1	0	0	0	0	0	0	0	PostgreSQL
84	71	10000	5	3	8	17	12	15	10	12	11	Health-ClosedPRs0009
85	89	6764	18	1	4	2	2	2	2	2	2	dconvert
86	96	86058	11	2	9	2	8	8	6	8	9	SS-X
87	82	20000	9	3	4	5	4	7	4	3	7	pom3c
88	70	4653	38	1	5	7	9	7	5	3	6	SQL_AllMeasurements
89	82	65536	16	2	5	5	5	6	5	5	5	SS-W
90	82	86058	12	1	3	3	2	2	2	2	2	exastencils
91	78	20000	9	3	5	7	4	5	5	5	4	pom3b
92	80	20000	9	3	6	4	4	5	4	5	4	pom3a
93	95	68640	14	1	1	1	1	0	1	0	1	7z
94	72	10127	19	4	16	12	3	6	10	9	8	BankChurners
95	62	53662	17	1	4	4	3	4	6	5	5	x264
96	58	53662	17	2	3	1	2	4	3	4	3	SS-N
97	80	10000	24	4	5	4	4	3	5	5	4	xomo_osp2
98	81	10000	24	4	3	3	3	4	4	4	4	xomo_osp
99	81	10000	24	4	5	3	4	3	4	5	4	xomo_flight
100	92	21759	35	3	12	10	13	14	11	13	12	Data_COVID19_Indonesia
101	79	10000	88	3	3	3	3	2	3	3	3	billing10k
102	93	166975	35	1	0	0	0	0	0	0	0	javagc
103	84	10000	124	3	4	4	3	3	3	3	4	Scrum10k
104	86	10000	128	3	6	4	4	5	5	4	4	FFM-125-25-0.50-SAT-1
105	81	100000	124	3	2	4	3	3	3	3	3	Scrum100k
106	85	10000	256	3	3	3	3	3	3	3	3	FFM-250-50-0.50-SAT-1
107	83	10000	510	3	4	3	4	4	3	3	3	FFM-500-100-0.50-SAT-1
108	85	10000	511	3	5	4	5	3	4	4	3	FM-500-100-1.00-SAT-1
109	85	10000	511	3	5	5	5	4	6	3	4	FM-500-100-0.75-SAT-1
110	81	10000	517	3	3	3	3	3	2	3	3	FM-500-100-0.25-SAT-1
111	82	10000	513	3	5	4	4	4	3	4	4	FM-500-100-0.50-SAT-1

Budget = 24 evaluations

SD of $d2h$ at budget 24 — all optimizers vs. all datasets

ID	Δ	Rows	|X|	|Y|	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
1	95	259	4	2	12	3	10	5	7	8	8	SS-H
2	90	196	3	2	3	4	3	4	4	4	4	SS-D
3	90	350	20	1	11	6	5	6	4	6	5	socks
4	92	192	9	1	3	2	4	1	1	3	1	Apache_AllMeasurements
5	93	206	3	2	8	13	10	10	8	9	11	SS-B
6	97	247	22	2	10	13	13	12	14	11	12	wallpaper
7	96	196	3	1	4	2	0	0	0	2	4	wc+rs-3d-c4-obj1
8	97	196	3	1	4	0	0	0	0	1	3	wc+sol-3d-c4-obj1
9	90	398	4	3	7	7	6	5	5	7	5	auto93
10	87	81	26	1	9	6	6	4	4	5	4	player_statistics_cleaned_final
11	91	196	3	2	3	4	4	3	5	3	4	SS-G
12	96	196	3	1	5	0	0	0	3	2	4	wc+wc-3d-c4-obj1
13	96	1152	16	1	1	1	2	0	1	1	1	X264_AllMeasurements
14	84	223	10	3	5	6	6	4	4	5	4	A2C_Acrobot
15	95	1023	11	2	11	5	11	7	4	7	7	SS-P
16	98	864	18	1	0	0	0	1	1	0	0	HSQLDB
17	95	93	22	3	5	0	2	6	4	2	4	nasa93dem
18	92	1024	10	1	10	6	8	3	3	6	7	LLVM
19	92	196	3	2	3	4	3	3	4	3	4	SS-F
20	82	972	11	2	5	4	4	5	3	3	3	SS-O
21	95	1023	11	2	11	5	11	6	7	10	7	SS-L
22	83	1343	3	2	13	2	2	2	2	2	2	SS-A
23	97	864	17	3	1	0	0	0	0	0	0	SS-M
24	97	649	33	1	44	0	0	0	31	0	0	student_dropout
25	81	1000	20	5	4	5	5	5	7	4	5	coc1000
26	79	1599	10	2	5	5	3	3	6	4	4	Wine_quality
27	93	2866	6	1	0	0	0	0	0	0	0	sol-6d-c2-obj1
28	89	2880	6	2	12	8	9	6	12	9	8	SS-K
29	78	1470	32	3	7	7	5	7	9	7	9	WA_Fn-UseC_-HR-Employee-Attrition
30	100	1557	12	1	0	0	0	0	0	0	0	storm
31	94	2880	6	1	3	0	1	1	1	1	2	wc-6d-c1-obj1
32	86	1000	124	3	3	4	5	3	4	3	4	Scrum1k
33	76	3008	14	2	2	2	3	6	3	1	1	SS-R
34	99	2560	16	1	0	0	0	0	0	0	0	BDBC
35	97	3840	6	1	3	1	0	0	1	0	0	rs-6d-c3_obj1
36	91	3840	6	2	13	7	5	6	18	5	5	SS-J
37	92	2736	13	3	4	4	4	8	1	2	2	SS-Q
38	100	3840	6	1	0	0	0	0	0	0	0	rs-6d-c3_obj2
39	95	3456	14	1	1	0	0	0	1	1	1	HSMGP_num
40	92	3840	6	2	1	1	1	2	2	1	1	SS-S
41	92	4096	12	1	3	4	5	2	3	2	3	deeparch
42	49	10000	5	3	3	4	2	7	4	4	3	Health-ClosedIssues0001
43	98	10000	5	3	0	38	0	0	0	0	0	Health-ClosedPRs0005
44	96	10000	5	3	22	0	0	0	0	14	0	Health-ClosedPRs0007
45	81	10000	5	3	10	5	2	5	7	6	7	Health-ClosedIssues0008
46	98	10000	5	3	0	18	0	0	0	0	0	Health-ClosedPRs0011
47	95	1460	77	4	9	8	7	6	8	4	5	home_data_for_ml_course
48	47	10000	5	3	12	8	31	10	0	5	5	Health-Commits0002
49	81	10000	5	3	7	6	5	7	6	6	6	Health-Commits0008
50	60	10000	5	3	0	4	4	5	4	4	4	Health-Commits0006
51	97	10000	5	3	0	0	22	1	5	0	0	Health-ClosedPRs0002
52	93	10000	5	3	24	21	17	13	14	17	19	Health-ClosedPRs0004
53	97	5160	9	1	40	15	0	0	26	0	0	dataset600
54	76	10000	5	3	14	14	14	11	12	11	13	Health-ClosedPRs0008
55	97	10000	5	3	19	19	0	0	0	0	0	Health-ClosedIssues0011
56	84	2938	20	2	4	7	4	6	4	5	4	Life_Expectancy_Data
57	81	10000	5	3	8	14	5	10	8	9	7	Health-ClosedPRs0010
58	43	10000	5	3	2	2	1	3	3	2	2	Health-ClosedIssues0004
59	67	10000	5	3	10	4	2	2	2	3	3	Health-Commits0010
60	77	10000	5	3	1	2	1	1	2	2	2	Health-Commits0009
61	91	5184	12	2	2	2	2	2	1	1	1	SS-T
62	60	4400	9	1	7	8	6	9	7	6	7	redis
63	80	10000	5	3	26	13	3	4	6	3	0	Health-ClosedIssues0002
64	98	10000	5	3	30	1	0	29	30	29	29	Health-ClosedPRs0006
65	88	10000	5	3	6	5	6	11	9	8	8	Health-Commits0003
66	95	10000	5	3	34	0	1	0	0	0	0	Health-Commits0001
67	75	10000	5	3	7	3	5	5	6	5	5	Health-ClosedIssues0003
68	79	10000	5	3	9	9	8	9	8	7	8	Health-ClosedIssues0009
69	28	10000	5	3	0	0	0	3	1	0	0	Health-Commits0004
70	81	10000	5	3	11	3	2	4	1	2	2	Health-ClosedIssues0006
71	33	10000	5	3	7	1	1	3	3	2	2	Health-ClosedIssues0010
72	97	10000	5	3	1	21	0	0	0	0	0	Health-ClosedPRs0003
73	72	10000	5	3	18	15	20	22	20	21	18	Health-ClosedIssues0007
74	90	6840	16	2	7	4	6	13	9	7	9	SS-V
75	61	10000	5	3	4	14	1	3	1	3	1	Health-ClosedIssues0005

Continued. . .

Budget 24 — continued

ID	Δ	Rows	|X|	|Y|	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
76	81	2205	31	8	3	2	3	5	4	4	4	Marketing_Analytics
77	74	10000	5	3	9	5	5	4	8	5	4	Health-Commits0000
78	89	4608	21	2	3	3	4	4	1	1	1	SS-U
79	98	10000	5	2	50	0	0	0	0	0	0	Health-ClosedPRs0000
80	91	10000	5	3	7	11	2	4	18	6	12	Health-Commits0007
81	97	10000	5	3	33	0	0	0	0	0	0	Health-Commits0011
82	92	7043	19	2	13	5	2	1	1	1	2	WA_Fn-UseC_-Telco-Customer-Churn
83	95	19008	9	1	0	0	0	0	0	0	0	PostgreSQL
84	71	10000	5	3	9	13	12	15	12	15	9	Health-ClosedPRs0009
85	89	6764	18	1	3	2	2	2	2	2	1	dconvert
86	96	86058	11	2	3	4	8	7	7	7	7	SS-X
87	82	20000	9	3	4	3	4	7	4	4	3	pom3c
88	70	4653	38	1	4	5	7	7	6	5	5	SQL_AllMeasurements
89	82	65536	16	2	5	5	5	5	6	6	7	SS-W
90	82	86058	12	1	3	2	2	2	2	1	3	exastencils
91	78	20000	9	3	4	6	3	5	5	4	5	pom3b
92	80	20000	9	3	5	6	4	4	5	4	5	pom3a
93	95	68640	14	1	1	1	1	1	0	0	1	7z
94	72	10127	19	4	16	11	3	4	6	6	8	BankChurners
95	62	53662	17	1	4	3	3	4	5	3	5	x264
96	58	53662	17	2	2	2	2	4	4	3	4	SS-N
97	80	10000	24	4	4	3	4	3	4	2	4	xomo_osp2
98	81	10000	24	4	2	3	3	4	3	4	4	xomo_osp
99	81	10000	24	4	5	3	4	3	3	3	3	xomo_flight
100	92	21759	35	3	11	9	13	14	14	11	12	Data_COVID19_Indonesia
101	79	10000	88	3	3	3	3	2	2	2	2	billing10k
102	93	166975	35	1	0	0	0	0	0	0	0	javagc
103	84	10000	124	3	3	3	3	3	4	3	4	Scrum10k
104	86	10000	128	3	5	4	4	4	3	3	4	FFM-125-25-0.50-SAT-1
105	81	100000	124	3	3	3	3	3	3	2	3	Scrum100k
106	85	10000	256	3	3	2	3	2	2	2	3	FFM-250-50-0.50-SAT-1
107	83	10000	510	3	4	5	3	4	3	4	3	FFM-500-100-0.50-SAT-1
108	85	10000	511	3	4	2	4	2	3	3	3	FM-500-100-1.00-SAT-1
109	85	10000	511	3	5	3	4	3	3	3	5	FM-500-100-0.75-SAT-1
110	81	10000	517	3	3	3	3	2	3	2	2	FM-500-100-0.25-SAT-1
111	82	10000	513	3	5	4	4	4	3	4	4	FM-500-100-0.50-SAT-1

Budget = 50 evaluations

SD of $d2h$ at budget 50 — all optimizers vs. all datasets

ID	Δ	Rows	|X|	|Y|	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
1	95	259	4	2	10	0	8	5	8	7	7	SS-H
2	90	196	3	2	2	2	3	3	5	1	2	SS-D
3	90	350	20	1	8	2	4	6	0	5	6	socks
4	92	192	9	1	2	1	2	0	1	1	2	Apache_AllMeasurements
5	93	206	3	2	9	5	9	10	0	9	8	SS-B
6	97	247	22	2	17	13	6	9	9	12	12	wallpaper
7	96	196	3	1	0	0	0	0	0	0	0	wc+rs-3d-c4-obj1
8	97	196	3	1	2	0	0	0	0	0	0	wc+sol-3d-c4-obj1
9	90	398	4	3	7	7	7	3	5	6	6	auto93
10	87	81	26	1	7	5	4	3	3	4	4	player_statistics_cleaned_final
11	91	196	3	2	3	3	2	3	4	3	3	SS-G
12	96	196	3	1	1	0	0	0	0	0	0	wc+wc-3d-c4-obj1
13	96	1152	16	1	1	1	1	0	0	1	1	X264_AllMeasurements
14	84	223	10	3	5	4	5	5	6	5	5	A2C_Acrobot
15	95	1023	11	2	1	1	7	1	1	4	5	SS-P
16	98	864	18	1	0	0	0	0	0	0	0	HSQLDB
17	95	93	22	3	2	5	3	3	3	0	2	nasa93dem
18	92	1024	10	1	2	1	6	0	2	3	2	LLVM
19	92	196	3	2	2	1	2	3	4	3	3	SS-F
20	82	972	11	2	4	4	3	4	2	4	3	SS-O
21	95	1023	11	2	1	1	7	4	1	2	5	SS-L
22	83	1343	3	2	2	1	2	1	1	1	2	SS-A
23	97	864	17	3	0	0	0	0	0	0	0	SS-M
24	97	649	33	1	37	0	0	0	0	0	0	student_dropout
25	81	1000	20	5	3	5	4	4	5	4	3	coc1000
26	79	1599	10	2	4	4	3	5	6	4	5	Wine_quality
27	93	2866	6	1	0	0	0	0	0	0	0	sol-6d-c2-obj1
28	89	2880	6	2	8	5	5	6	11	5	7	SS-K
29	78	1470	32	3	11	4	2	7	9	6	6	WA_Fn-UseC_-HR-Employee-Attrition
30	100	1557	12	1	0	0	0	0	0	0	0	storm
31	94	2880	6	1	2	0	0	0	2	0	1	wc-6d-c1-obj1
32	86	1000	124	3	3	3	4	2	3	2	3	Scrum1k
33	76	3008	14	2	1	2	2	4	2	1	1	SS-R
34	99	2560	16	1	0	0	0	0	0	0	0	BDBC
35	97	3840	6	1	2	0	0	0	1	0	0	rs-6d-c3_obj1
36	91	3840	6	2	9	1	1	1	1	2	2	SS-J
37	92	2736	13	3	3	0	2	0	1	1	1	SS-Q
38	100	3840	6	1	0	0	0	0	0	0	0	rs-6d-c3_obj2
39	95	3456	14	1	0	0	0	0	0	0	0	HSMGP_num
40	92	3840	6	2	2	1	1	2	1	1	1	SS-S
41	92	4096	12	1	0	0	2	0	2	3	0	deeparch
42	49	10000	5	3	15	3	2	8	6	5	5	Health-ClosedIssues0001
43	98	10000	5	3	0	17	0	0	0	0	0	Health-ClosedPRs0005
44	96	10000	5	3	0	0	0	0	14	0	0	Health-ClosedPRs0007
45	81	10000	5	3	9	3	2	4	3	3	5	Health-ClosedIssues0008
46	98	10000	5	3	0	0	0	0	0	0	0	Health-ClosedPRs0011
47	95	1460	77	4	16	4	7	6	8	7	5	home_data_for_ml_course
48	47	10000	5	3	14	8	4	11	5	10	1	Health-Commits0002
49	81	10000	5	3	9	7	6	7	6	6	6	Health-Commits0008
50	60	10000	5	3	10	4	4	4	3	4	4	Health-Commits0006
51	97	10000	5	3	0	0	0	2	1	0	0	Health-ClosedPRs0002
52	93	10000	5	3	22	14	7	13	0	0	0	Health-ClosedPRs0004
53	97	5160	9	1	40	18	0	0	0	0	0	dataset600
54	76	10000	5	3	15	12	9	7	10	7	7	Health-ClosedPRs0008
55	97	10000	5	3	1	26	0	0	0	0	0	Health-ClosedIssues0011
56	84	2938	20	2	6	7	3	4	2	5	3	Life_Expectancy_Data
57	81	10000	5	3	11	12	11	11	13	11	8	Health-ClosedPRs0010
58	43	10000	5	3	8	3	5	4	4	2	1	Health-ClosedIssues0004
59	67	10000	5	3	10	4	5	3	4	3	2	Health-Commits0010
60	77	10000	5	3	8	1	1	1	1	1	1	Health-Commits0009
61	91	5184	12	2	2	1	2	2	1	0	0	SS-T
62	60	4400	9	1	8	6	9	8	6	5	6	redis
63	80	10000	5	3	14	1	3	6	3	4	3	Health-ClosedIssues0002
64	98	10000	5	3	2	24	23	24	22	21	28	Health-ClosedPRs0006
65	88	10000	5	3	4	8	6	10	8	9	7	Health-Commits0003
66	95	10000	5	3	16	0	0	0	0	0	0	Health-Commits0001
67	75	10000	5	3	11	3	2	4	2	4	5	Health-ClosedIssues0003
68	79	10000	5	3	4	7	3	9	8	6	7	Health-ClosedIssues0009
69	28	10000	5	3	1	1	1	3	1	3	1	Health-Commits0004
70	81	10000	5	3	7	3	2	3	2	2	3	Health-ClosedIssues0006
71	33	10000	5	3	4	3	4	3	3	2	2	Health-ClosedIssues0010
72	97	10000	5	3	1	0	0	0	0	0	0	Health-ClosedPRs0003
73	72	10000	5	3	21	20	1	24	18	24	21	Health-ClosedIssues0007
74	90	6840	16	2	3	1	4	5	4	8	7	SS-V
75	61	10000	5	3	24	19	1	3	1	1	1	Health-ClosedIssues0005

Continued. . .

Budget 50 — continued

ID	Δ	Rows	|X|	|Y|	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
76	81	2205	31	8	2	2	3	5	5	3	4	Marketing_Analytics
77	74	10000	5	3	3	6	6	5	6	5	5	Health-Commits0000
78	89	4608	21	2	2	1	3	1	0	1	1	SS-U
79	98	10000	5	2	37	0	0	0	0	0	0	Health-ClosedPRs0000
80	91	10000	5	3	20	4	6	2	4	5	6	Health-Commits0007
81	97	10000	5	3	19	28	0	0	0	0	0	Health-Commits0011
82	92	7043	19	2	9	4	0	1	0	0	0	WA_Fn-UseC_-Telco-Customer-Churn
83	95	19008	9	1	0	0	0	0	0	0	0	PostgreSQL
84	71	10000	5	3	18	19	20	19	13	13	9	Health-ClosedPRs0009
85	89	6764	18	1	2	1	1	2	0	0	0	dconvert
86	96	86058	11	2	2	1	2	8	3	6	4	SS-X
87	82	20000	9	3	3	4	4	5	5	3	4	pom3c
88	70	4653	38	1	3	4	6	6	5	3	4	SQL_AllMeasurements
89	82	65536	16	2	6	5	4	5	4	5	6	SS-W
90	82	86058	12	1	2	2	2	1	1	1	1	exastencils
91	78	20000	9	3	3	2	2	4	3	3	3	pom3b
92	80	20000	9	3	4	3	3	3	3	4	3	pom3a
93	95	68640	14	1	1	1	1	1	0	0	0	7z
94	72	10127	19	4	17	12	2	5	4	4	3	BankChurners
95	62	53662	17	1	3	4	2	4	4	2	5	x264
96	58	53662	17	2	1	2	1	4	4	2	3	SS-N
97	80	10000	24	4	3	3	3	2	3	2	3	xomo_osp2
98	81	10000	24	4	3	2	2	2	1	2	3	xomo_osp
99	81	10000	24	4	4	3	3	2	1	2	2	xomo_flight
100	92	21759	35	3	8	11	8	13	6	8	9	Data_COVID19_Indonesia
101	79	10000	88	3	2	2	3	1	1	2	2	billing10k
102	93	166975	35	1	0	0	0	0	0	0	0	javagc
103	84	10000	124	3	2	2	3	2	3	3	3	Scrum10k
104	86	10000	128	3	5	3	3	2	3	5	4	FFM-125-25-0.50-SAT-1
105	81	100000	124	3	3	2	2	2	2	2	3	Scrum100k
106	85	10000	256	3	2	2	3	1	2	2	2	FFM-250-50-0.50-SAT-1
107	83	10000	510	3	2	1	3	1	3	2	2	FFM-500-100-0.50-SAT-1
108	85	10000	511	3	3	2	3	2	3	2	2	FM-500-100-1.00-SAT-1
109	85	10000	511	3	4	3	4	2	2	3	3	FM-500-100-0.75-SAT-1
110	81	10000	517	3	2	2	3	2	2	2	2	FM-500-100-0.25-SAT-1
111	82	10000	513	3	4	4	4	2	3	3	2	FM-500-100-0.50-SAT-1

Budget = 100 evaluations

SD of $d2h$ at budget 100 — all optimizers vs. all datasets

ID	Δ	Rows	|X|	|Y|	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
1	95	259	4	2	0	0	6	5	2	4	5	SS-H
2	90	196	3	2	2	1	2	1	1	0	2	SS-D
3	90	350	20	1	4	0	2	6	0	2	3	socks
4	92	192	9	1	1	1	1	0	1	1	1	Apache_AllMeasurements
5	93	206	3	2	9	0	9	5	0	7	6	SS-B
6	97	247	22	2	7	12	6	3	0	10	10	wallpaper
7	96	196	3	1	0	0	0	0	0	0	0	wc+rs-3d-c4-obj1
8	97	196	3	1	0	0	0	0	0	0	0	wc+sol-3d-c4-obj1
9	90	398	4	3	9	5	5	1	2	4	4	auto93
10	87	81	26	1	4	4	4	0	0	3	3	player_statistics_cleaned_final
11	91	196	3	2	1	0	2	3	2	2	2	SS-G
12	96	196	3	1	0	0	0	0	0	0	0	wc+wc-3d-c4-obj1
13	96	1152	16	1	1	0	0	0	0	0	0	X264_AllMeasurements
14	84	223	10	3	5	5	4	2	4	4	4	A2C_Acrobot
15	95	1023	11	2	1	0	4	0	1	2	1	SS-P
16	98	864	18	1	0	0	0	0	0	0	0	HSQLDB
17	95	93	22	3	0	1	2	0	1	0	2	nasa93dem
18	92	1024	10	1	1	1	5	0	2	2	1	LLVM
19	92	196	3	2	1	0	1	1	2	2	1	SS-F
20	82	972	11	2	4	3	4	3	1	2	2	SS-O
21	95	1023	11	2	1	0	4	1	0	1	1	SS-L
22	83	1343	3	2	2	0	1	1	1	2	2	SS-A
23	97	864	17	3	0	0	0	0	0	0	0	SS-M
24	97	649	33	1	51	0	0	0	0	0	0	student_dropout
25	81	1000	20	5	3	3	5	3	4	2	3	coc1000
26	79	1599	10	2	4	5	4	5	5	3	3	Wine_quality
27	93	2866	6	1	0	0	0	0	0	0	0	sol-6d-c2-obj1
28	89	2880	6	2	5	2	3	4	3	4	4	SS-K
29	78	1470	32	3	3	3	3	6	2	5	3	WA_Fn-UseC_-HR-Employee-Attrition
30	100	1557	12	1	0	0	0	0	0	0	0	storm
31	94	2880	6	1	1	0	0	0	0	0	0	wc-6d-c1-obj1
32	86	1000	124	3	3	2	4	2	2	2	2	Scrum1k
33	76	3008	14	2	1	2	2	3	2	2	2	SS-R
34	99	2560	16	1	0	0	0	0	0	0	0	BDBC
35	97	3840	6	1	1	0	0	0	0	0	0	rs-6d-c3_obj1
36	91	3840	6	2	5	1	1	1	1	1	2	SS-J
37	92	2736	13	3	1	0	0	0	0	0	0	SS-Q
38	100	3840	6	1	0	0	0	0	0	0	0	rs-6d-c3_obj2
39	95	3456	14	1	0	0	0	0	0	0	0	HSMGP_num
40	92	3840	6	2	1	0	0	1	1	1	1	SS-S
41	92	4096	12	1	0	0	2	0	0	0	0	deeparch
42	49	10000	5	3	13	4	7	8	8	7	5	Health-ClosedIssues0001
43	98	10000	5	3	0	0	39	0	0	0	0	Health-ClosedPRs0005
44	96	10000	5	3	14	0	0	0	0	0	0	Health-ClosedPRs0007
45	81	10000	5	3	10	2	5	2	3	3	2	Health-ClosedIssues0008
46	98	10000	5	3	0	0	0	0	0	0	0	Health-ClosedPRs0011
47	95	1460	77	4	6	8	7	7	7	6	8	home_data_for_ml_course
48	47	10000	5	3	24	19	32	11	11	14	8	Health-Commits0002
49	81	10000	5	3	9	5	6	8	5	4	5	Health-Commits0008
50	60	10000	5	3	30	4	3	5	5	4	4	Health-Commits0006
51	97	10000	5	3	0	0	0	4	5	0	0	Health-ClosedPRs0002
52	93	10000	5	3	20	17	0	14	0	0	0	Health-ClosedPRs0004
53	97	5160	9	1	34	0	0	0	0	0	0	dataset600
54	76	10000	5	3	15	9	12	7	1	1	2	Health-ClosedPRs0008
55	97	10000	5	3	0	0	0	0	0	0	0	Health-ClosedIssues0011
56	84	2938	20	2	9	5	3	2	1	4	3	Life_Expectancy_Data
57	81	10000	5	3	8	10	12	12	12	9	8	Health-ClosedPRs0010
58	43	10000	5	3	3	2	3	4	4	3	2	Health-ClosedIssues0004
59	67	10000	5	3	17	4	2	4	2	3	3	Health-Commits0010
60	77	10000	5	3	10	1	1	1	1	1	1	Health-Commits0009
61	91	5184	12	2	1	1	1	1	1	0	0	SS-T
62	60	4400	9	1	6	7	8	7	7	8	8	redis
63	80	10000	5	3	15	4	4	5	6	5	3	Health-ClosedIssues0002
64	98	10000	5	3	22	26	29	21	13	18	22	Health-ClosedPRs0006
65	88	10000	5	3	3	8	2	8	6	8	7	Health-Commits0003
66	95	10000	5	3	3	0	11	0	0	0	0	Health-Commits0001
67	75	10000	5	3	7	3	2	3	2	3	3	Health-ClosedIssues0003
68	79	10000	5	3	4	4	6	7	8	5	8	Health-ClosedIssues0009
69	28	10000	5	3	10	5	1	5	6	3	0	Health-Commits0004
70	81	10000	5	3	5	3	3	3	2	2	3	Health-ClosedIssues0006
71	33	10000	5	3	3	3	4	3	3	2	2	Health-ClosedIssues0010
72	97	10000	5	3	16	0	15	0	0	0	0	Health-ClosedPRs0003
73	72	10000	5	3	1	23	18	23	24	24	23	Health-ClosedIssues0007
74	90	6840	16	2	1	1	2	1	4	3	7	SS-V
75	61	10000	5	3	18	1	1	3	4	1	1	Health-ClosedIssues0005

Continued...

Budget 100 — continued

ID	Δ	Rows	$\|X\|$	$\|Y\|$	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
76	81	2205	31	8	2	2	2	4	3	3	3	Marketing_Analytics
77	74	10000	5	3	6	4	5	6	5	6	5	Health-Commits0000
78	89	4608	21	2	2	2	3	0	0	1	0	SS-U
79	98	10000	5	2	47	0	0	0	0	0	0	Health-ClosedPRs0000
80	91	10000	5	3	22	2	5	1	1	4	1	Health-Commits0007
81	97	10000	5	3	15	14	0	0	0	0	0	Health-Commits0011
82	92	7043	19	2	13	1	1	1	0	0	0	WA_Fn-UseC_-Telco-Customer-Churn
83	95	19008	9	1	0	0	0	0	0	0	0	PostgreSQL
84	71	10000	5	3	20	17	15	22	16	16	10	Health-ClosedPRs0009
85	89	6764	18	1	0	1	1	1	0	0	0	dconvert
86	96	86058	11	2	1	0	1	7	3	6	6	SS-X
87	82	20000	9	3	3	4	4	4	4	4	3	pom3c
88	70	4653	38	1	2	3	6	7	4	3	4	SQL_AllMeasurements
89	82	65536	16	2	5	5	4	5	4	5	5	SS-W
90	82	86058	12	1	1	1	1	1	1	1	1	exastencils
91	78	20000	9	3	3	2	2	2	2	2	3	pom3b
92	80	20000	9	3	3	3	2	2	2	3	2	pom3a
93	95	68640	14	1	0	0	0	1	0	0	0	7z
94	72	10127	19	4	12	7	5	5	4	2	2	BankChurners
95	62	53662	17	1	2	3	2	4	3	3	3	x264
96	58	53662	17	2	1	2	1	4	2	3	2	SS-N
97	80	10000	24	4	3	3	3	2	2	2	3	xomo_osp2
98	81	10000	24	4	2	2	2	1	1	2	2	xomo_osp
99	81	10000	24	4	4	2	1	1	1	2	2	xomo_flight
100	92	21759	35	3	5	9	8	14	4	6	5	Data_COVID19_Indonesia
101	79	10000	88	3	2	2	2	1	1	1	2	billing10k
102	93	166975	35	1	0	0	0	0	0	0	0	javagc
103	84	10000	124	3	2	3	2	1	3	2	2	Scrum10k
104	86	10000	128	3	3	2	4	1	2	2	3	FFM-125-25-0.50-SAT-1
105	81	100000	124	3	2	2	2	2	2	2	2	Scrum100k
106	85	10000	256	3	1	2	2	1	1	2	1	FFM-250-50-0.50-SAT-1
107	83	10000	510	3	2	2	3	1	2	3	3	FFM-500-100-0.50-SAT-1
108	85	10000	511	3	1	2	2	1	2	2	2	FM-500-100-1.00-SAT-1
109	85	10000	511	3	2	3	3	1	3	3	3	FM-500-100-0.75-SAT-1
110	81	10000	517	3	2	3	3	1	2	2	2	FM-500-100-0.25-SAT-1
111	82	10000	513	3	3	3	4	2	2	3	2	FM-500-100-0.50-SAT-1

Budget = 200 evaluations

SD of $d2h$ at budget 200 — all optimizers vs. all datasets

ID	Δ	Rows	|X|	|Y|	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
1	95	259	4	2	0	0	0	0	0	0	0	SS-H
2	90	196	3	2	2	0	2	0	0	0	0	SS-D
3	90	350	20	1	3	0	0	2	0	0	0	socks
4	92	192	9	1	0	0	1	0	0	0	1	Apache_AllMeasurements
5	93	206	3	2	6	0	6	0	0	1	3	SS-B
6	97	247	22	2	3	9	3	0	0	5	7	wallpaper
7	96	196	3	1	0	0	0	0	0	0	0	wc+rs-3d-c4-obj1
8	97	196	3	1	0	0	0	0	0	0	0	wc+sol-3d-c4-obj1
9	90	398	4	3	5	5	5	0	0	3	2	auto93
10	87	81	26	1	3	3	1	0	0	3	2	player_statistics_cleaned_final
11	91	196	3	2	0	0	1	0	0	0	1	SS-G
12	96	196	3	1	0	0	0	0	0	0	0	wc+wc-3d-c4-obj1
13	96	1152	16	1	0	0	0	0	0	0	0	X264_AllMeasurements
14	84	223	10	3	4	5	4	0	0	3	4	A2C_Acrobot
15	95	1023	11	2	0	0	1	0	1	1	1	SS-P
16	98	864	18	1	0	0	0	0	0	0	0	HSQLDB
17	95	93	22	3	8	1	0	0	1	0	1	nasa93dem
18	92	1024	10	1	0	0	4	0	1	1	2	LLVM
19	92	196	3	2	0	0	0	0	0	0	1	SS-F
20	82	972	11	2	3	1	4	2	0	1	2	SS-O
21	95	1023	11	2	0	0	1	1	0	1	1	SS-L
22	83	1343	3	2	1	0	1	0	0	1	2	SS-A
23	97	864	17	3	0	0	0	0	0	0	0	SS-M
24	97	649	33	1	0	0	0	0	0	0	0	student_dropout
25	81	1000	20	5	2	3	4	2	2	2	3	coc1000
26	79	1599	10	2	5	4	3	4	2	3	3	Wine_quality
27	93	2866	6	1	0	0	0	0	0	0	0	sol-6d-c2-obj1
28	89	2880	6	2	5	1	3	2	1	3	2	SS-K
29	78	1470	32	3	2	1	2	4	2	5	5	WA_Fn-UseC_-HR-Employee-Attrition
30	100	1557	12	1	0	0	0	0	0	0	0	storm
31	94	2880	6	1	1	0	0	0	0	0	0	wc-6d-c1-obj1
32	86	1000	124	3	2	2	3	1	2	2	2	Scrum1k
33	76	3008	14	2	2	2	2	1	2	2	2	SS-R
34	99	2560	16	1	0	0	0	0	0	0	0	BDBC
35	97	3840	6	1	0	0	0	0	0	0	0	rs-6d-c3_obj1
36	91	3840	6	2	2	1	1	1	1	1	1	SS-J
37	92	2736	13	3	0	0	0	0	0	0	0	SS-Q
38	100	3840	6	1	0	0	0	0	0	0	0	rs-6d-c3_obj2
39	95	3456	14	1	0	0	0	0	0	0	0	HSMGP_num
40	92	3840	6	2	1	0	0	1	0	0	0	SS-S
41	92	4096	12	1	0	0	0	0	0	0	0	deeparch
42	49	10000	5	3	13	7	5	9	5	5	5	Health-ClosedIssues0001
43	98	10000	5	3	40	0	27	0	0	0	0	Health-ClosedPRs0005
44	96	10000	5	3	26	0	0	0	0	0	0	Health-ClosedPRs0007
45	81	10000	5	3	12	2	2	3	3	3	2	Health-ClosedIssues0008
46	98	10000	5	3	16	35	32	0	0	0	0	Health-ClosedPRs0011
47	95	1460	77	4	5	11	8	8	3	9	9	home_data_for_ml_course
48	47	10000	5	3	3	18	7	12	15	14	14	Health-Commits0002
49	81	10000	5	3	7	2	5	7	2	6	2	Health-Commits0008
50	60	10000	5	3	14	3	3	3	4	4	4	Health-Commits0006
51	97	10000	5	3	0	0	0	2	2	0	0	Health-ClosedPRs0002
52	93	10000	5	3	12	10	0	11	0	0	0	Health-ClosedPRs0004
53	97	5160	9	1	44	0	0	0	0	0	0	dataset600
54	76	10000	5	3	14	3	15	7	3	2	1	Health-ClosedPRs0008
55	97	10000	5	3	36	0	33	0	0	0	0	Health-ClosedIssues0011
56	84	2938	20	2	7	7	3	1	1	2	2	Life_Expectancy_Data
57	81	10000	5	3	14	5	10	16	6	8	9	Health-ClosedPRs0010
58	43	10000	5	3	7	4	6	5	4	2	4	Health-ClosedIssues0004
59	67	10000	5	3	11	4	3	3	3	3	3	Health-Commits0010
60	77	10000	5	3	10	1	1	1	1	0	1	Health-Commits0009
61	91	5184	12	2	1	0	1	1	0	0	0	SS-T
62	60	4400	9	1	6	8	9	7	5	6	6	redis
63	80	10000	5	3	4	1	5	5	5	5	5	Health-ClosedIssues0002
64	98	10000	5	3	1	0	18	18	0	0	0	Health-ClosedPRs0006
65	88	10000	5	3	19	4	1	7	5	6	6	Health-Commits0003
66	95	10000	5	3	33	0	0	1	0	0	0	Health-Commits0001
67	75	10000	5	3	12	3	2	2	3	3	2	Health-ClosedIssues0003
68	79	10000	5	3	13	10	10	3	5	7	7	Health-ClosedIssues0009
69	28	10000	5	3	7	6	0	9	7	3	3	Health-Commits0004
70	81	10000	5	3	2	1	2	3	1	2	1	Health-ClosedIssues0006
71	33	10000	5	3	7	4	1	3	3	6	3	Health-ClosedIssues0010
72	97	10000	5	3	22	0	0	0	0	0	0	Health-ClosedPRs0003
73	72	10000	5	3	1	23	1	22	25	20	21	Health-ClosedIssues0007
74	90	6840	16	2	1	1	2	1	1	2	2	SS-V
75	61	10000	5	3	3	1	6	5	1	1	3	Health-ClosedIssues0005

Continued. . .

Budget 200 — continued

ID	Δ	Rows	$\|X\|$	$\|Y\|$	DEHB	SMAC	TPE	LITE	EZR	LINE	Random	Dataset
76	81	2205	31	8	2	1	2	2	2	3	3	Marketing_Analytics
77	74	10000	5	3	11	4	5	7	6	5	5	Health-Commits0000
78	89	4608	21	2	1	1	2	0	0	0	0	SS-U
79	98	10000	5	2	0	0	0	0	0	0	0	Health-ClosedPRs0000
80	91	10000	5	3	22	5	4	1	1	1	1	Health-Commits0007
81	97	10000	5	3	20	0	0	0	0	0	0	Health-Commits0011
82	92	7043	19	2	12	3	0	1	0	0	0	WA_Fn-UseC_-Telco-Customer-Churn
83	95	19008	9	1	0	0	0	0	0	0	0	PostgreSQL
84	71	10000	5	3	16	16	19	25	15	13	15	Health-ClosedPRs0009
85	89	6764	18	1	0	0	0	0	0	0	0	dconvert
86	96	86058	11	2	0	0	1	8	6	5	3	SS-X
87	82	20000	9	3	4	4	3	5	4	3	3	pom3c
88	70	4653	38	1	2	3	4	6	3	3	2	SQL_AllMeasurements
89	82	65536	16	2	5	5	5	5	3	4	4	SS-W
90	82	86058	12	1	1	0	1	1	1	1	1	exastencils
91	78	20000	9	3	2	2	1	1	1	2	2	pom3b
92	80	20000	9	3	2	1	2	2	1	1	2	pom3a
93	95	68640	14	1	0	0	0	0	0	0	0	7z
94	72	10127	19	4	10	2	1	5	3	5	4	BankChurners
95	62	53662	17	1	3	2	2	4	3	4	3	x264
96	58	53662	17	2	1	1	1	4	4	3	2	SS-N
97	80	10000	24	4	2	3	2	1	1	2	2	xomo_osp2
98	81	10000	24	4	1	1	1	1	1	2	1	xomo_osp
99	81	10000	24	4	1	2	1	1	1	1	1	xomo_flight
100	92	21759	35	3	10	7	6	13	3	8	6	Data_COVID19_Indonesia
101	79	10000	88	3	1	2	1	1	1	2	1	billing10k
102	93	166975	35	1	0	0	0	0	0	0	0	javagc
103	84	10000	124	3	1	3	2	1	2	3	2	Scrum10k
104	86	10000	128	3	2	3	3	0	3	3	3	FFM-125-25-0.50-SAT-1
105	81	100000	124	3	1	2	2	1	2	2	2	Scrum100k
106	85	10000	256	3	1	2	2	1	1	1	1	FFM-250-50-0.50-SAT-1
107	83	10000	510	3	1	2	3	2	1	2	2	FFM-500-100-0.50-SAT-1
108	85	10000	511	3	1	2	2	0	2	2	2	FM-500-100-1.00-SAT-1
109	85	10000	511	3	0	3	2	1	2	2	3	FM-500-100-0.75-SAT-1
110	81	10000	517	3	1	2	3	1	1	2	1	FM-500-100-0.25-SAT-1
111	82	10000	513	3	1	3	3	2	3	3	2	FM-500-100-0.50-SAT-1